

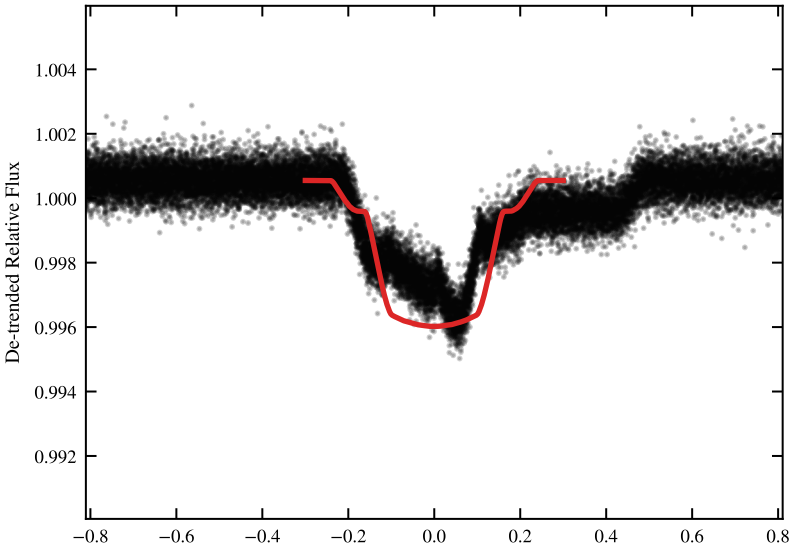
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 117789567 | RA: 300.77402° | Dec: 26.89203° | Sectors: 54, 55

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	2.3202 ± 0.000145
Rp/R★	0.06882 ± 0.0103
Rp (R⊕)	35.311 ± 12.3
b	0.76202 ± 0.22
T14 (h)	7.7814 ± 1.64
a (AU)	0.043488 ± 0.0151
Teq (K)	6881.7 ± 507

Primary Phased Transit



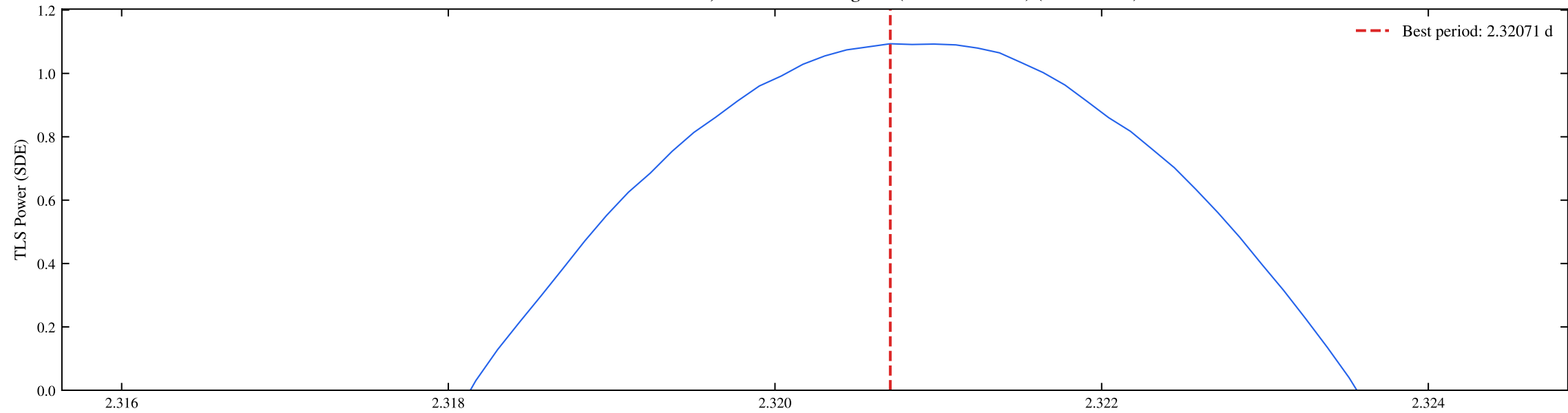
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	2.32071	2.320188 ± 0.00015
P1 Epoch (BTJD)	N/A	2770.2735 ± 0.1202
Max R-hat (R)	N/A	2.4105
Min ESS	N/A	5
MCMC Divergences	N/A	0

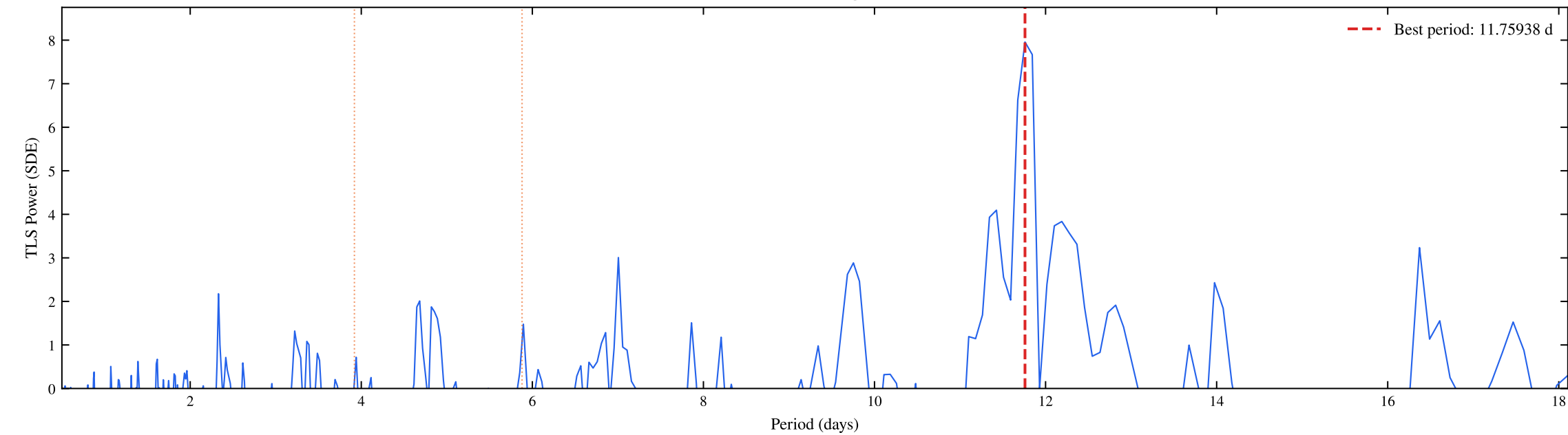
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	5.134 ± 1.54	Derived (Teff scaling)
M★ (M☉)	4.012	Torres 2010
Teff (K)	1.308e+04	VizieR TIC8
log g (dex)	3.235 ± 0.0161	Gaia DR3
[Fe/H]	-0.9989 ± 0.00325	Gaia DR3
ρ★ (g/cm³)	0.04179	Derived

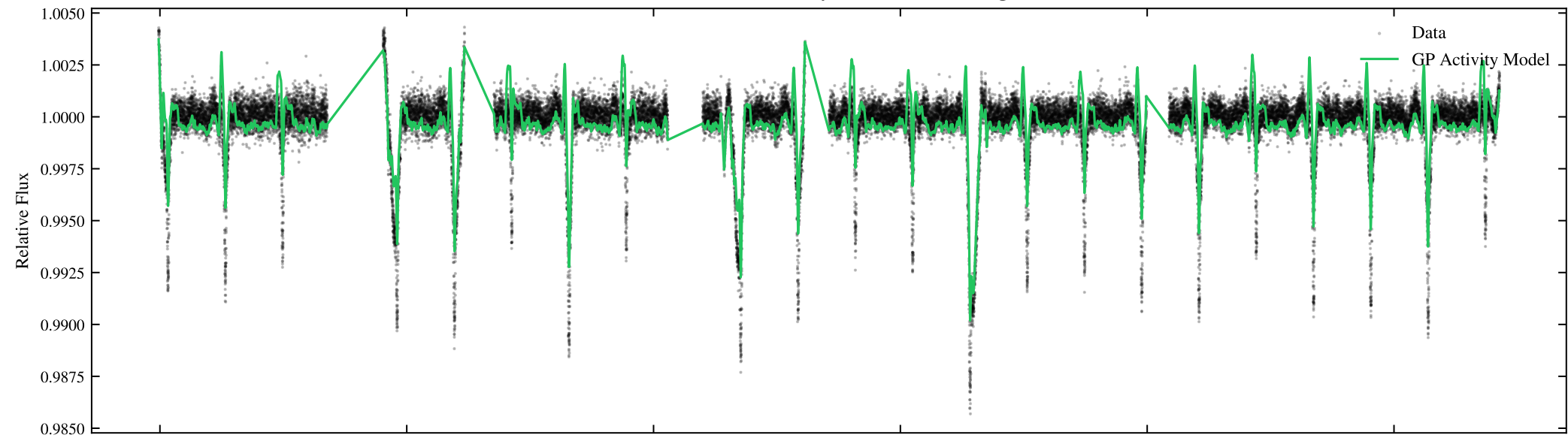
TIC 117789567 | Sectors: 54, 55 | TLS Periodogram (Planet 1 Search) (SDE = 1.09)



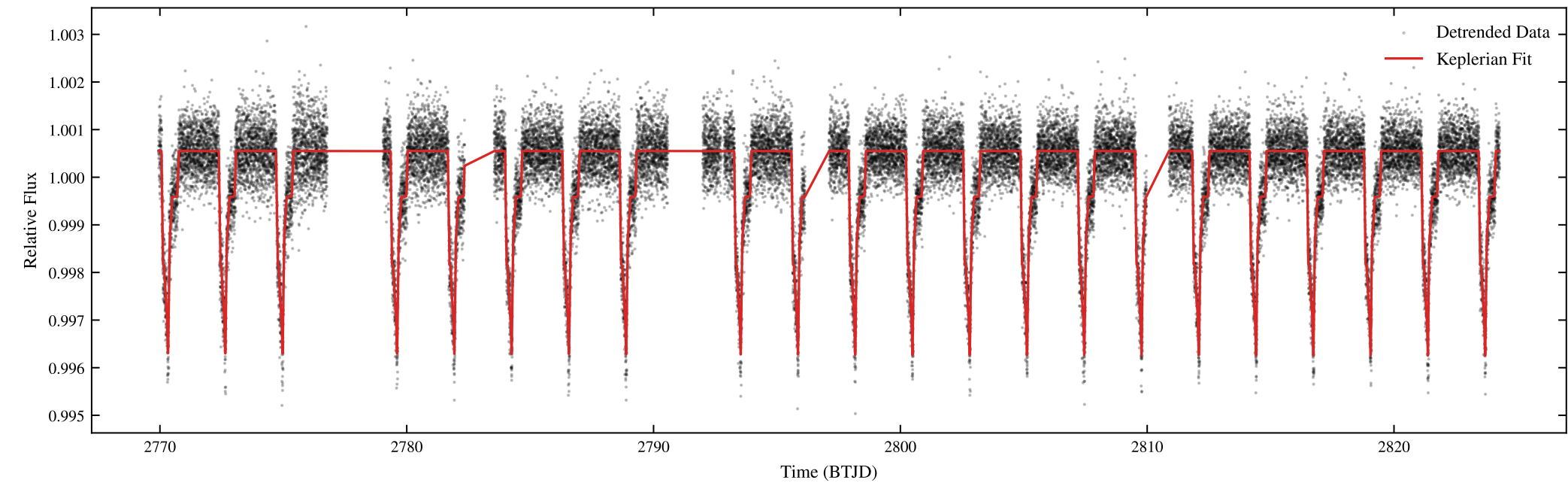
TIC 117789567 | Sectors: 54, 55 | TLS Periodogram (Planet 2 Search) (SDE = 7.96)



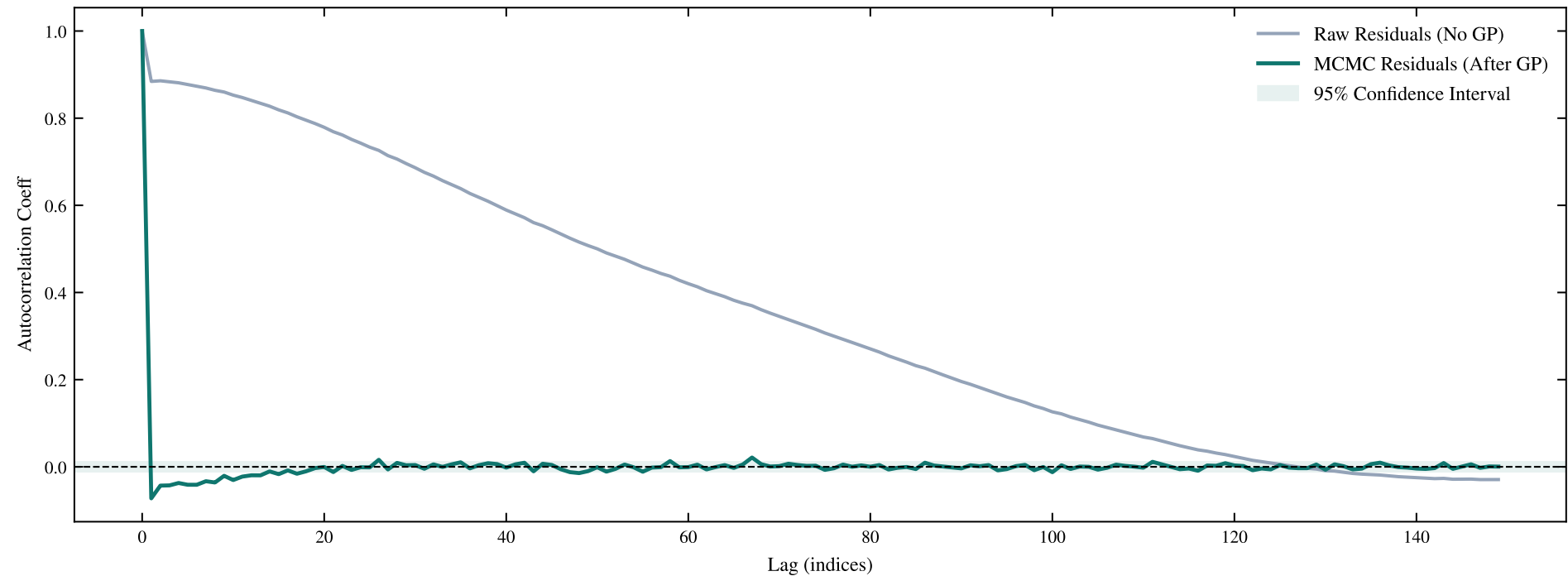
TIC 117789567 | Sectors: 54, 55 | Bayesian GP Detrending (Time Domain)



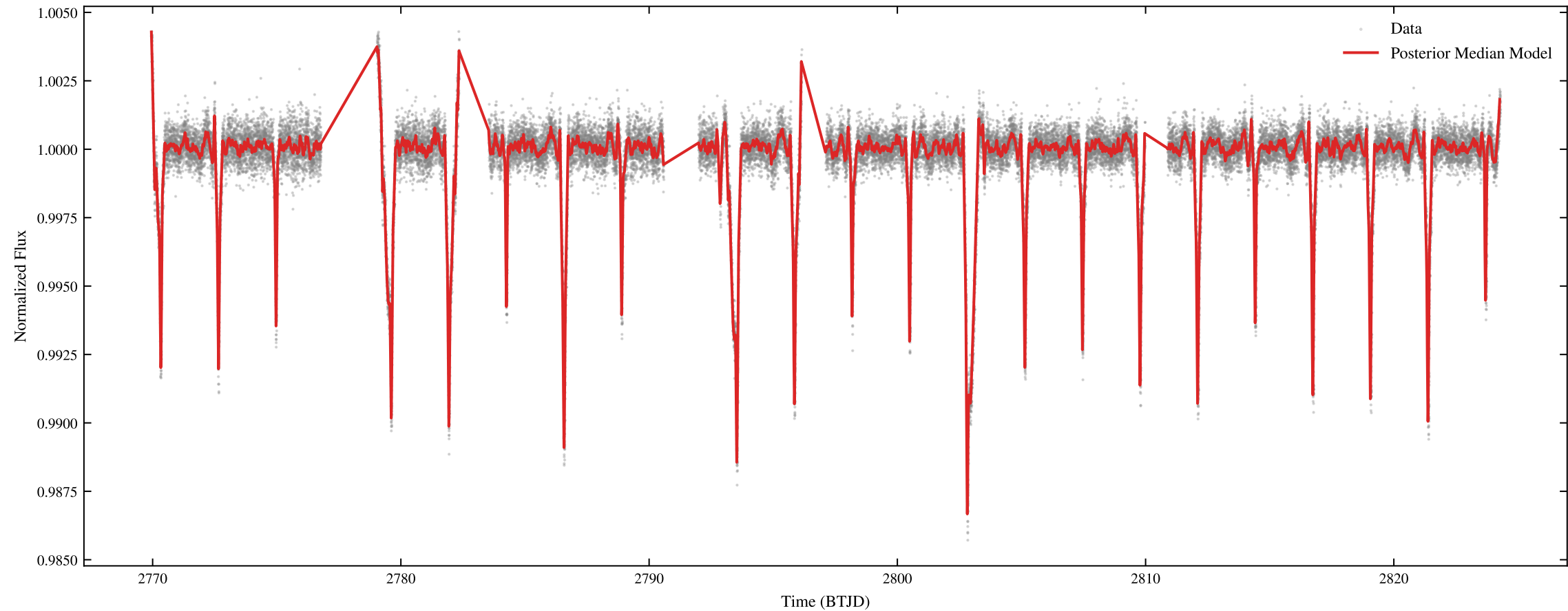
Detrended Light Curve & Best-Fit Transit Model

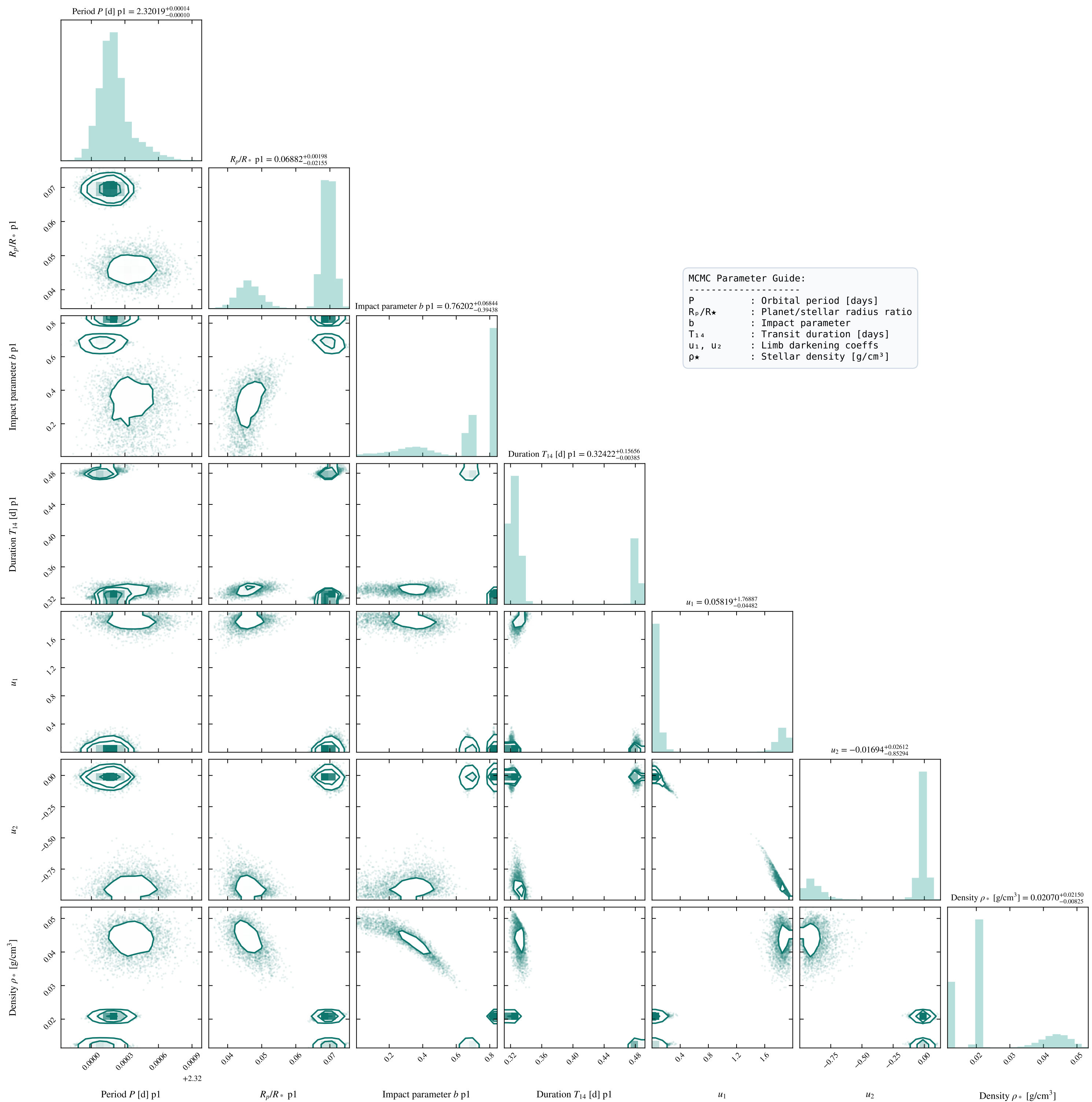


TIC 117789567 | Sectors: 54, 55 | GP Residuals Autocorrelation Function (ACF)

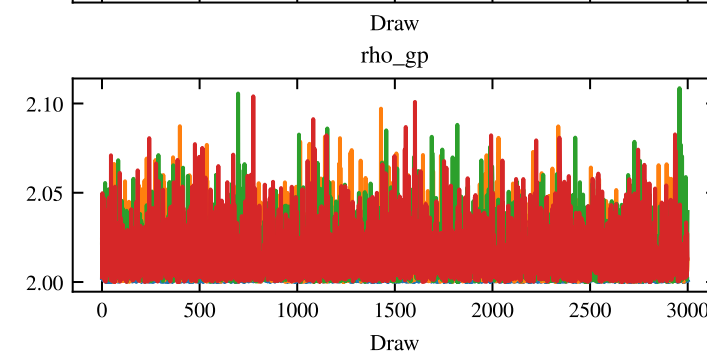
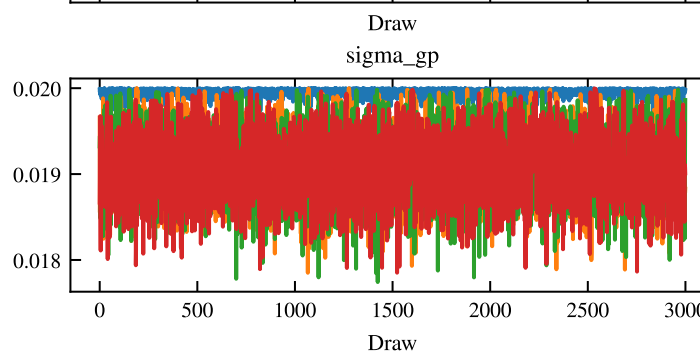
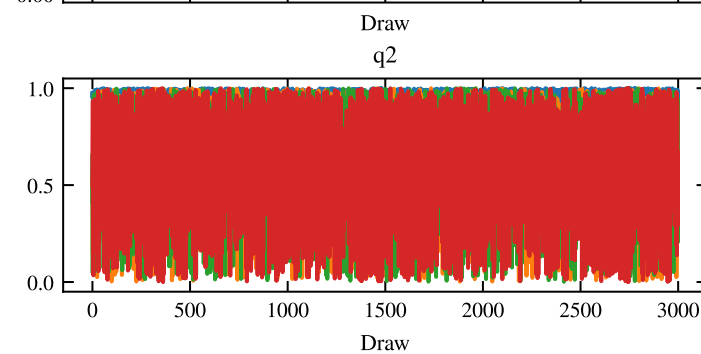
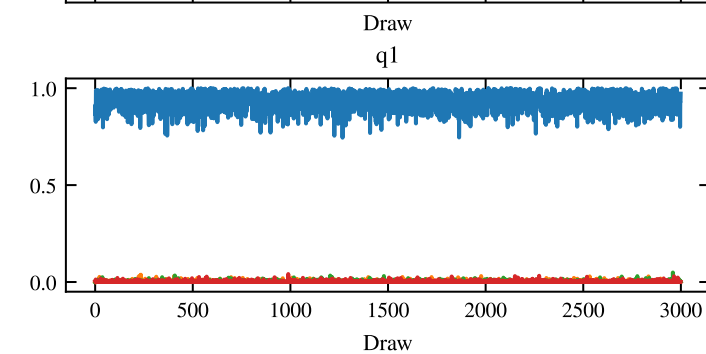
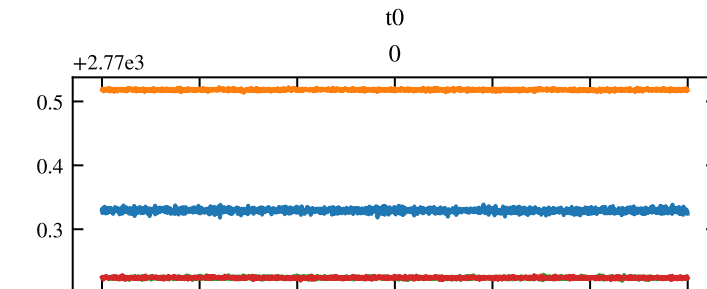
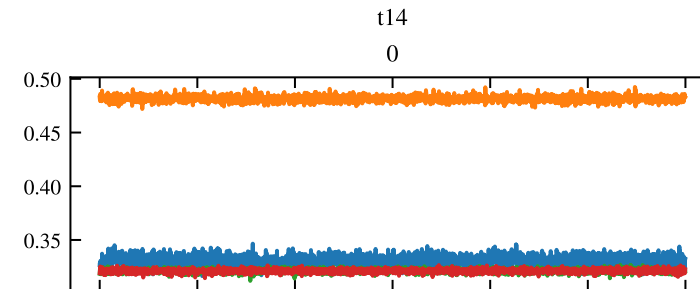
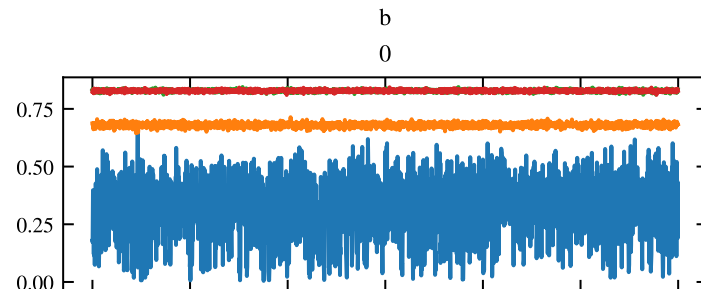
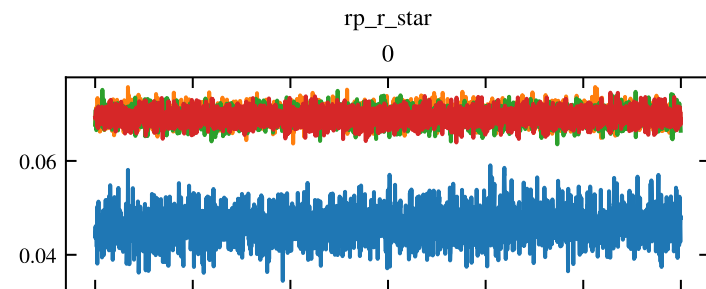


TIC 117789567 | Sectors: 54, 55 | MCMC Posterior Predictive Check

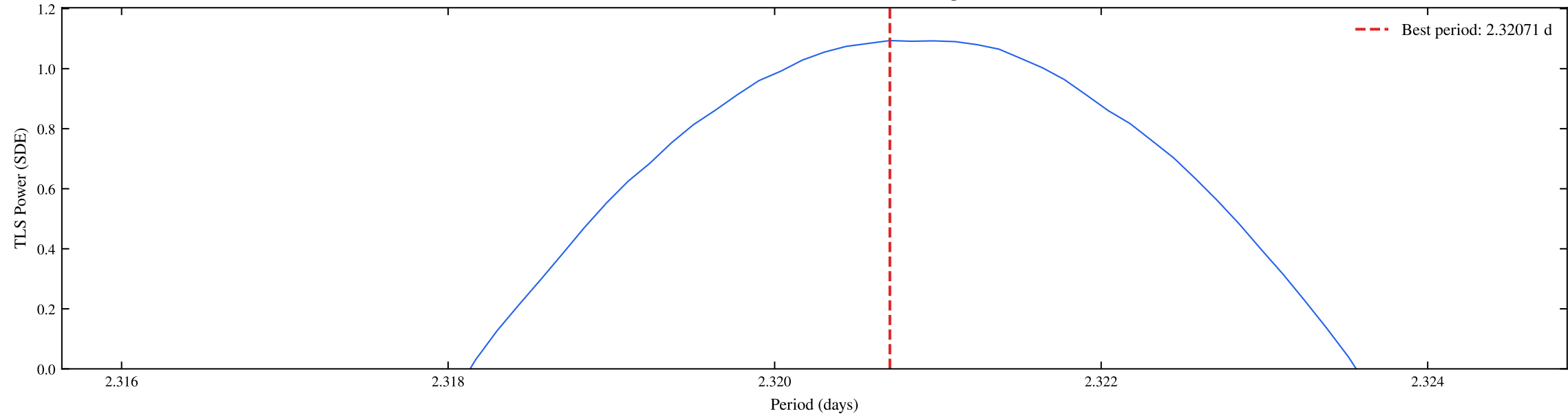




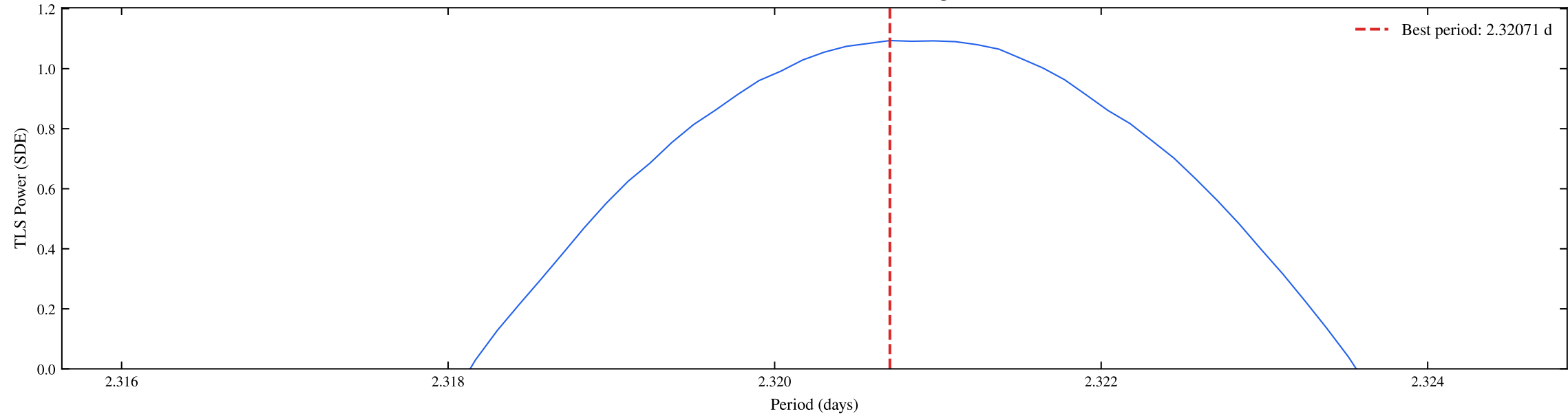
TIC 117789567 | Sectors: 54, 55 | MCMC Trace Plots



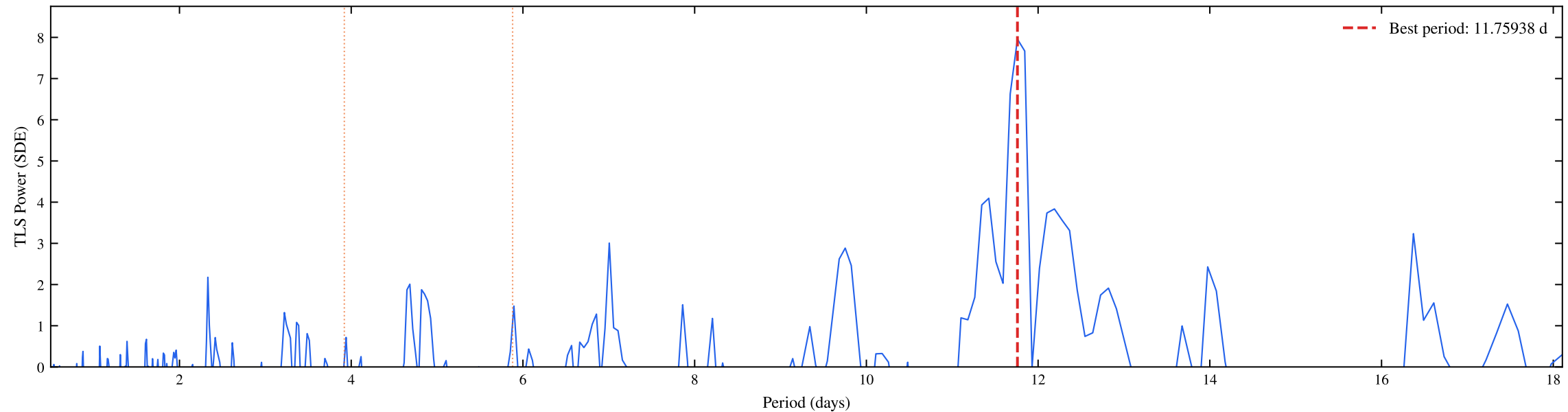
TIC 117789567 | Sectors: 54, 55 | TLS Coarse Periodogram (SDE = 1.09)



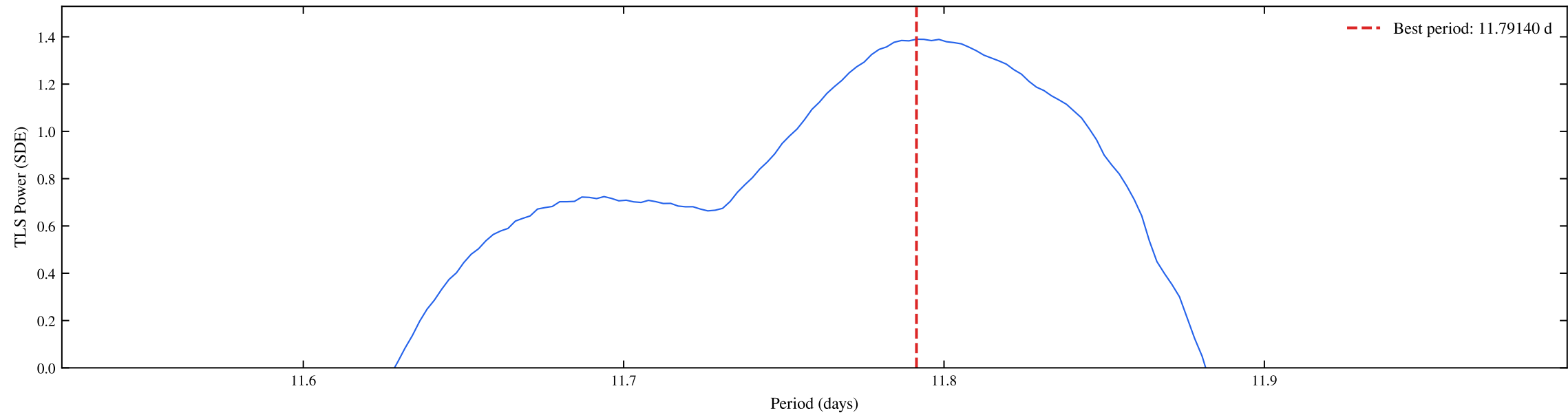
TIC 117789567 | Sectors: 54, 55 | TLS Fine Periodogram (SDE = 1.09)



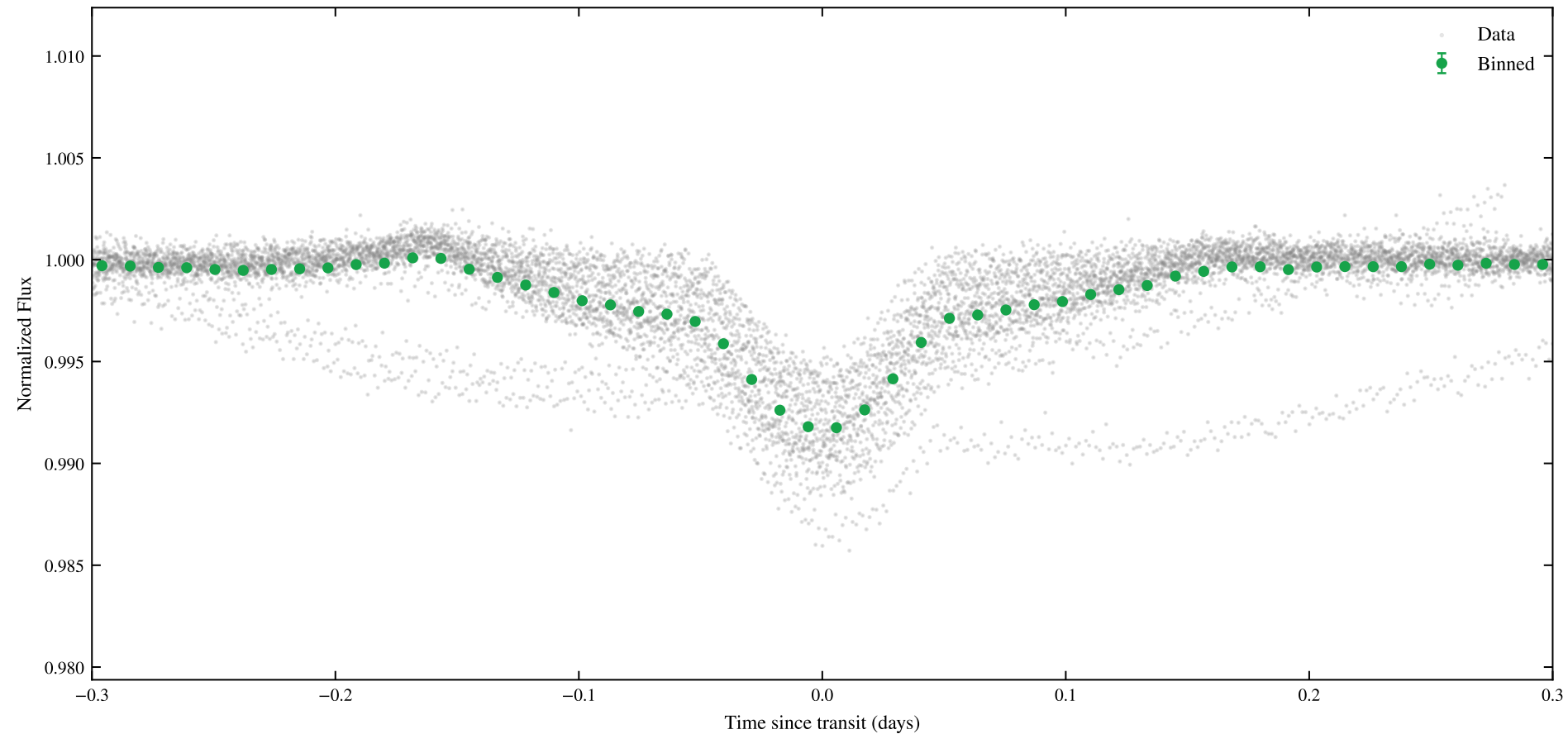
TIC 117789567 | Sectors: 54, 55 | TLS Coarse Periodogram (SDE = 7.96)



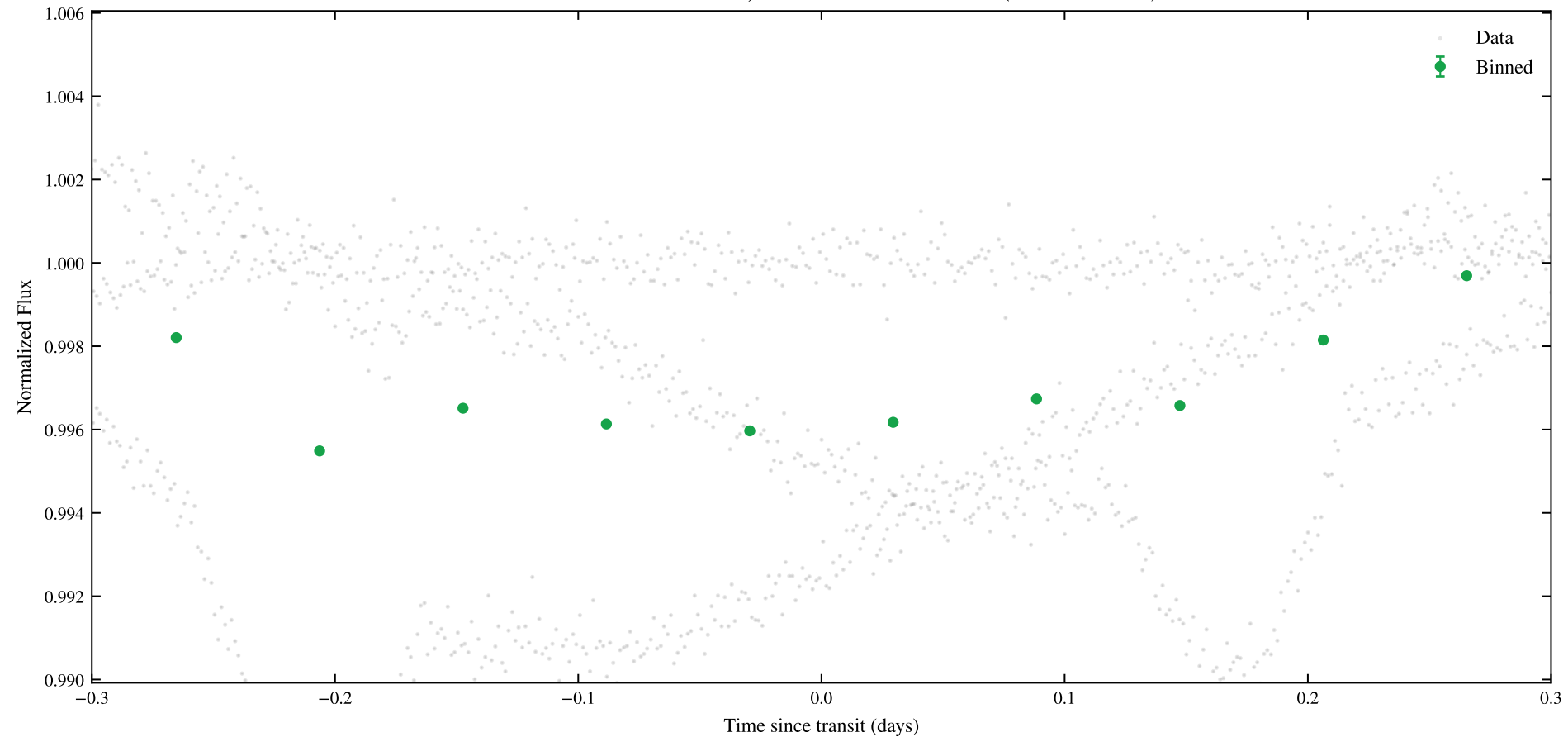
TIC 117789567 | Sectors: 54, 55 | TLS Fine Periodogram (SDE = 1.39)



TIC 117789567 | Sectors: 54, 55 | Phase-Folded Transit ($P = 2.32071$ d)



TIC 117789567 | Sectors: 54, 55 | Phase-Folded Transit (P = 11.79140 d)



TIC 117789567 | Sectors: 54, 55 | Stacked Individual Transits

